

Tracing Moral Foundations in Large Language Models



Chenxiao Yu*, Bowen Yi*, Shrikanth Narayanan, Morteza Dehghani, et al
Department of Computer Science, University of Southern California

Introduction

Problem:

LLMs generate human-like moral judgments, but is this genuine internal structure or just superficial "moral mimicry"?

Theory:

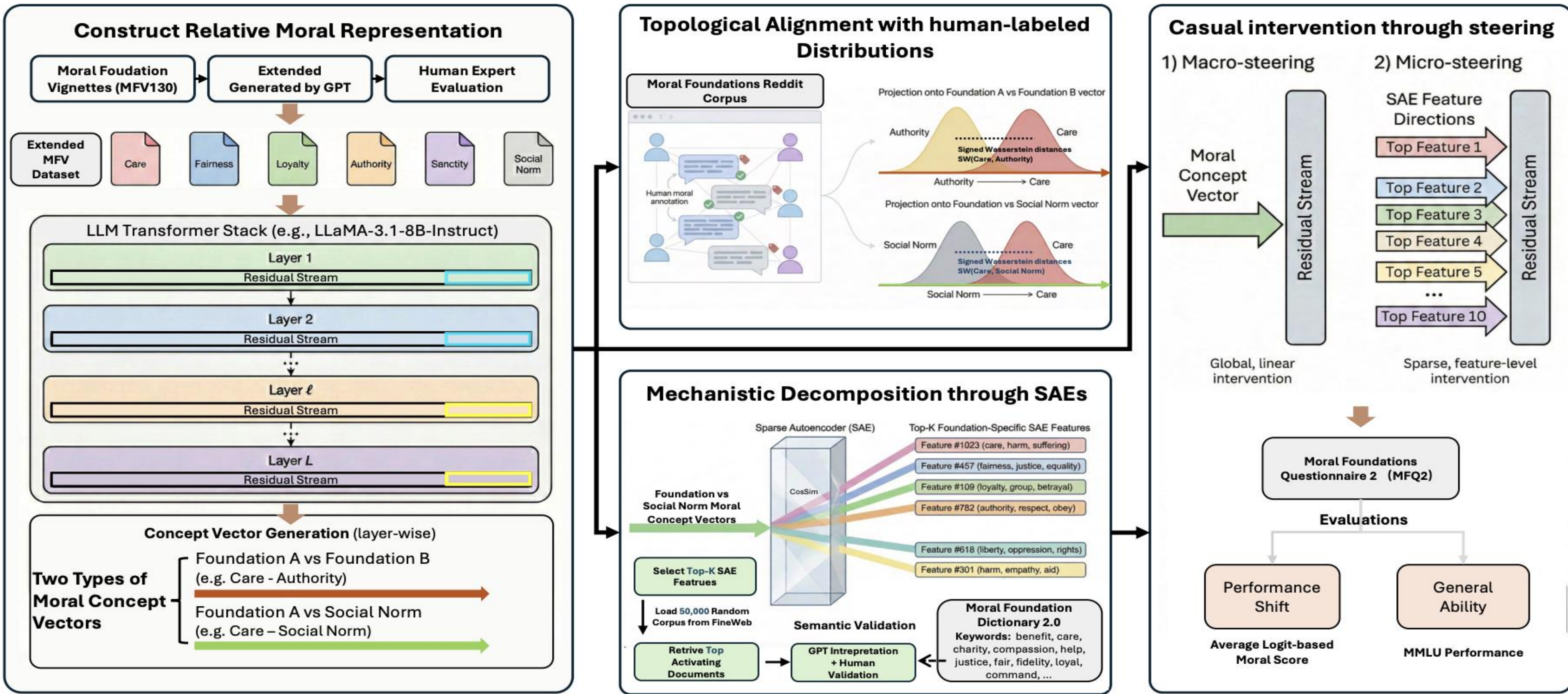
Moral Foundations Theory (MFT) is our analytical framework. MFT posits that human morality isn't good or bad, but has five dimensions: **Care, Fairness, Loyalty, Authority, and Sanctity**.

We investigate how LLMs (instruction-tuned **Llama-3.1-8B** and **Qwen-2.5-7B**) encode, organize, and apply these five moral foundation concepts.

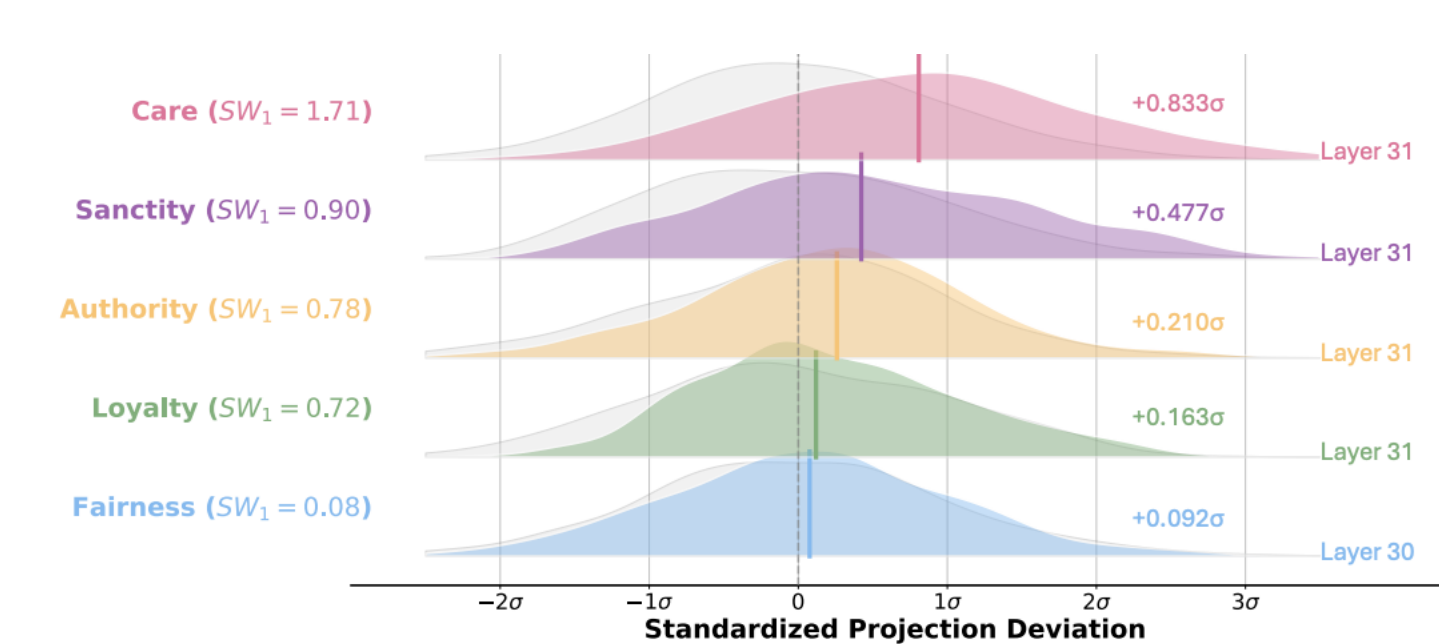
Research Questions & Contributions:

- Topological Alignment:** We find that the model's internal clustering of moral concepts matches the patterns in human-labeled data.
- Mechanistic Decomposition:** Using **Sparse Autoencoders (SAEs)**, we decompose abstract moral foundations into interpretable features (e.g. "Care" foundation splits into distinct components for "physical suffering" versus "emotional distress").
- Causal Control:** By intervening on these internal directions, we can predictably shift the model's downstream moral judgments, proving that these structures are causal drivers of behavior rather than just correlations.
- Support for Moral Pluralism:** We identify specific linear directions for each foundation. This geometric separability supports the theory that morality is multi-dimensional (pluralist) rather than single-dimensional.

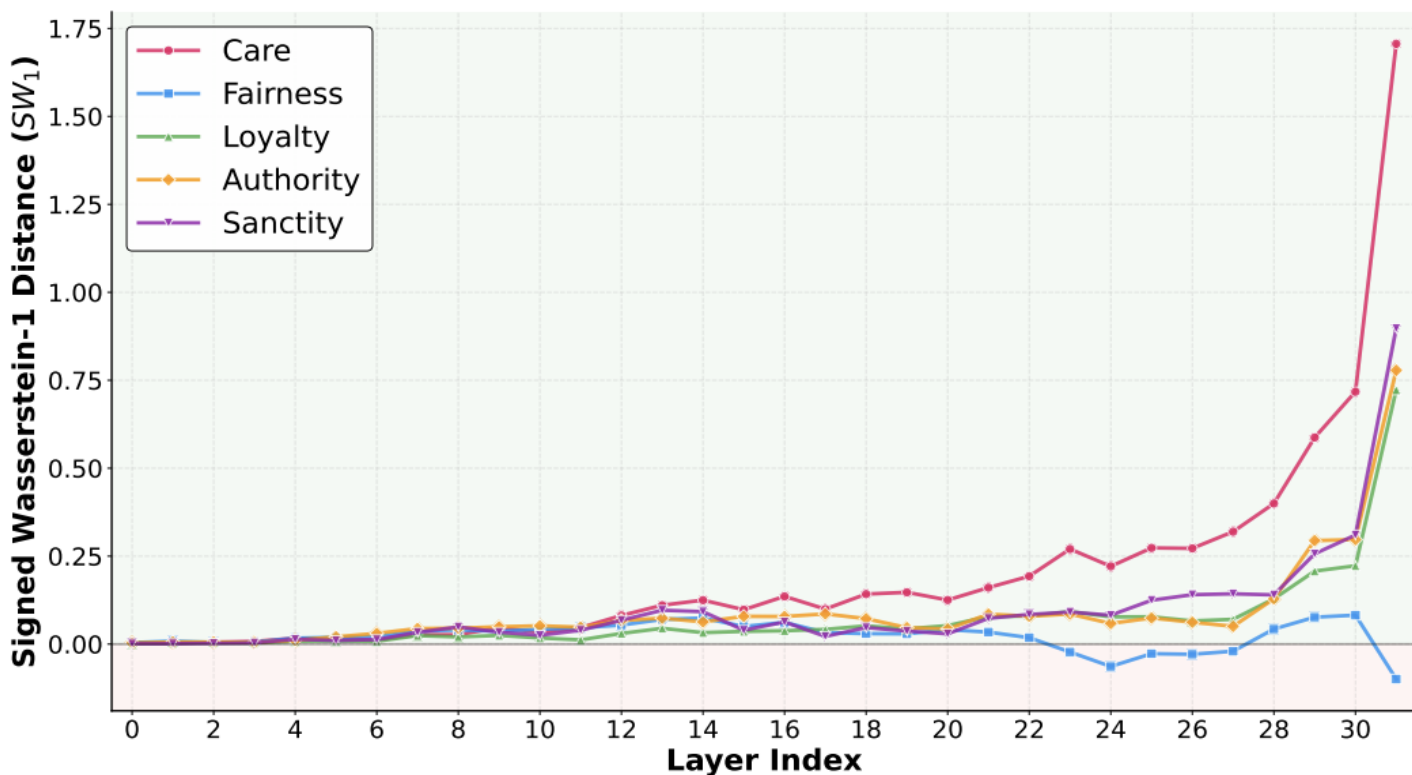
Methodology



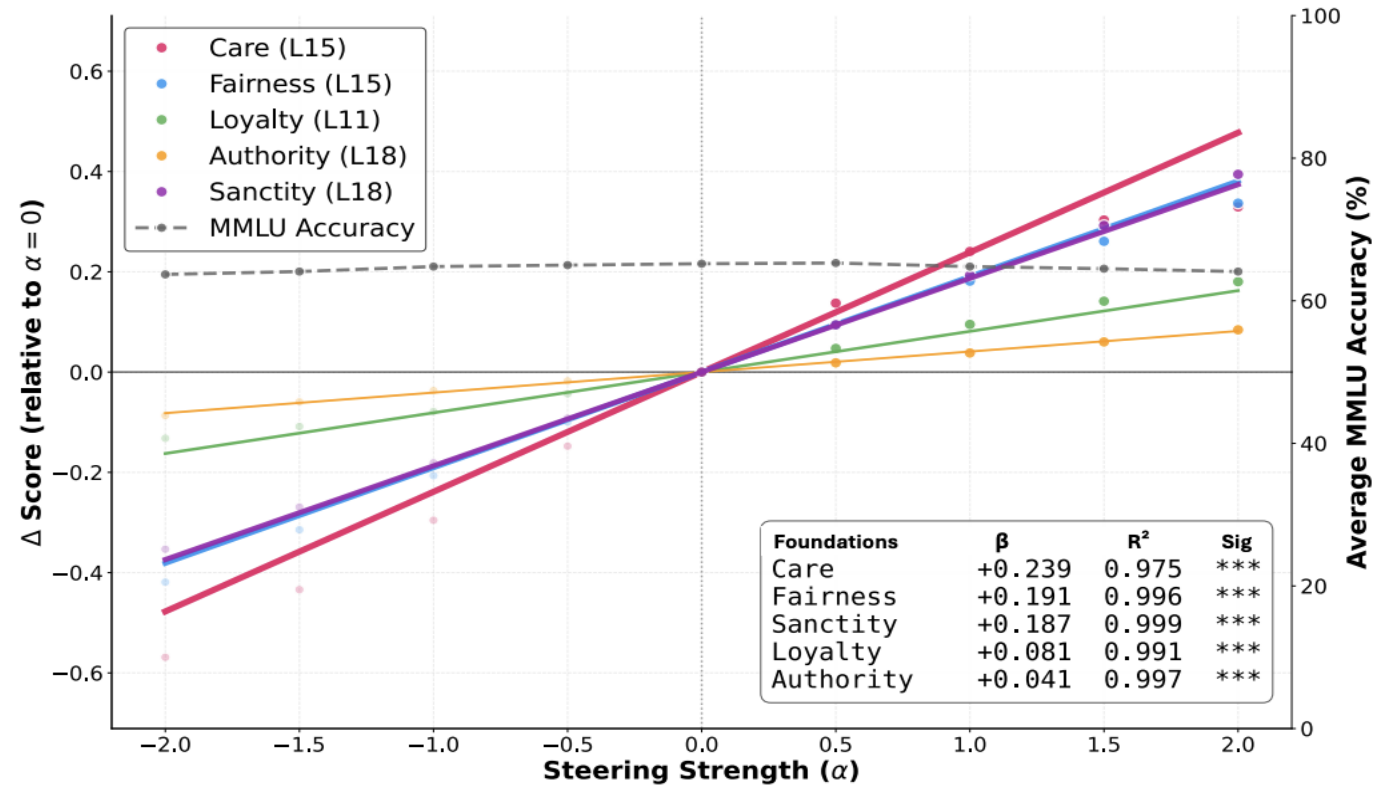
Result (More on paper)



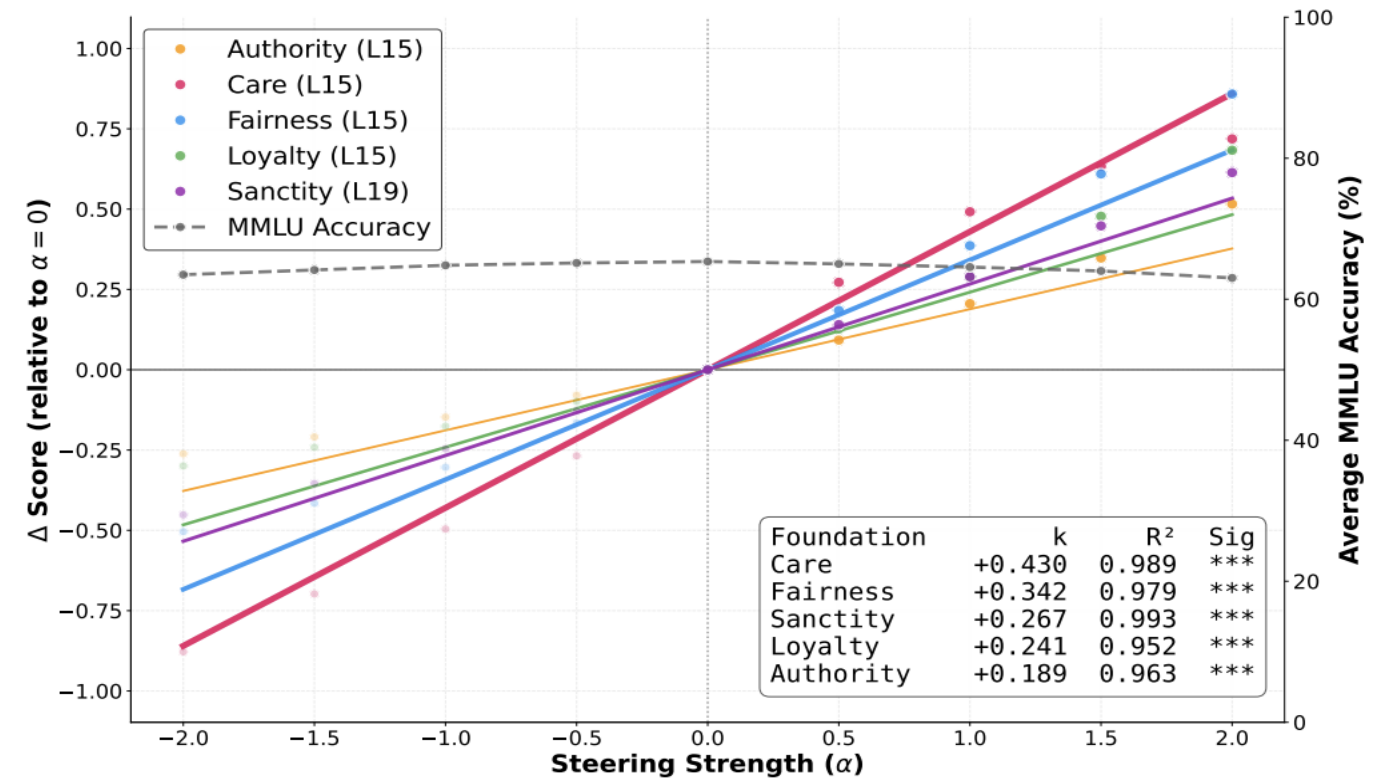
(a) **Topological Separation at the Boundary.** Standardized probability densities of moral projections (colored) vs. the non-moral baseline (gray). The x-axis represents semantic shift in standard deviations (σ).



(b) **Layer-wise Evolution of Separability.** Signed Wasserstein-1 distances (SW_1) across all 32 layers, where green indicates positive separability and red indicates negative (reversed) separability.



(a) **Macro (vector) steering response curves.** Steering along foundation-level concept vectors.



(b) **Micro (SAE) steering result.** Steering at Top-10 SAE feature at optimal layers.

Table: GPT-5.1 grounded, human validated semantic characterization of sample SAE features based on highly activated text for **Llama**. Complete tables for both LLMs please see the paper.

Layer	Feature ID	MFT	Conf.	Label (GPT, human-validated)	Peak-centered sample text (trimmed)
23	44965	Care	0.93	Graphic descriptions of extreme suffering and atrocity	Fly strike kills thousands of rabbits. ... The eggs hatch into maggots which eat their way into the poor rabbit's flesh. The rabbit dies from being eaten alive—a slow, painfully horrific death ...
11	8003	Authority	0.90	Polite , expert-style, authoritative explanatory answers to questions	... he only needs to be organized ... Please share any suggestions ... Ah, organization ... Dealing with clutter and putting things in order is an issue for almost everyone ...
15	41465	Care	0.90	Descriptions of large-scale suffering and atrocities	... the Holocaust, concentration camps, Nazi unfathomable brutality ...
19	90260	Authority	0.90	Organizational management systems, methods, and processes	... improvements in fields such as ... safe minimum levels of maintenance ... operating procedures and strategies ... capital maintenance regimes and plans ...
23	90226	Authority	0.86	Definitions and hierarchies of roles and positions	When thinking of a company organizational chart ... All positions ultimately lead up to an executive member. The executive is considered the leader of the company